

Date 2080122124
Class BSc 2nd Year
Roll No 68
Shift Day:

Physics Practical Sheet

Prithvi Narayan Campus

Experiment No 04
Group Physics
Sub Physics
Set

Object of the Experiment (Block Letters) : TO CONSTRUCT AND VERIFY 'NOR' AND 'NAND' GATE USING TTL (TRANSISTOR - TRANSISTOR LOGIC)

APPARATUS REQUIRED:

- (i) Transistors (ii) Wires (iii) Input Voltage
(iv) Multimeter (v) Bread board

THEORY:

(I) NOR GATE:

NOR gate is the combination of NOT and OR gate. It behaves according to the truth table. A high output results if both the inputs to the gate are low and if one or both input is high, a low output result. NOR is the result of the negation of the OR operator.

Boolean expression $\Rightarrow Y = \overline{A+B}$

Truth Table

Input		Output
A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

(II) NAND GATE:

NAND gate is a type of gate that is formed by the combination of a AND and NOT gate i.e. if AND gate output is given to a NOT gate, the combination forms NAND gate. The output due

② NOR Gate

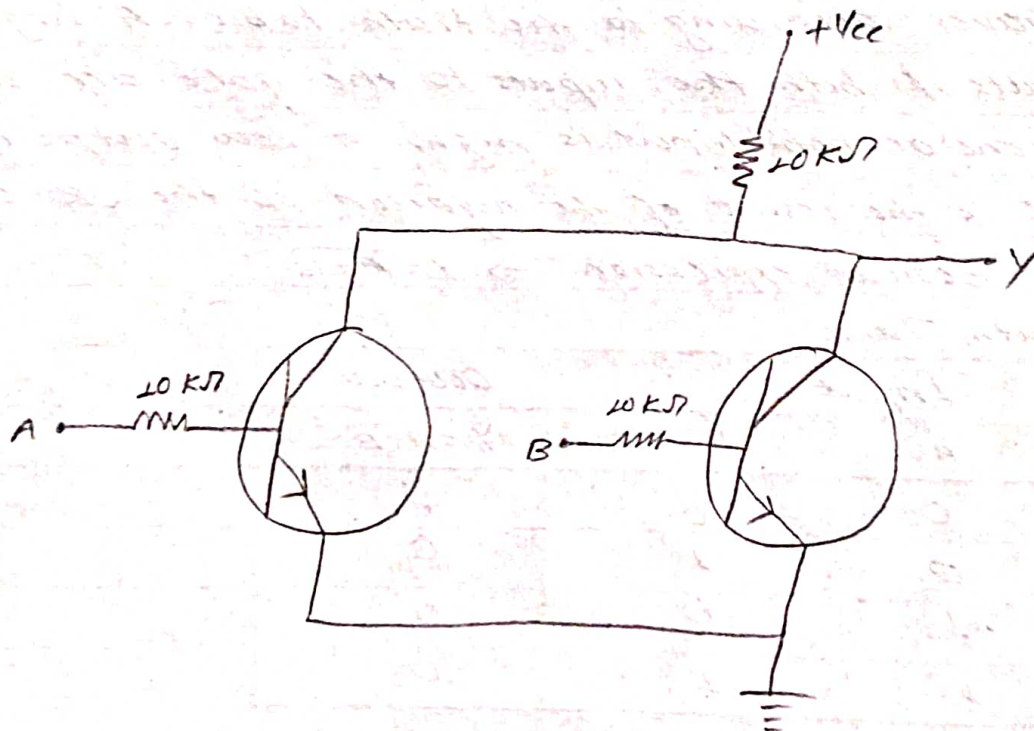
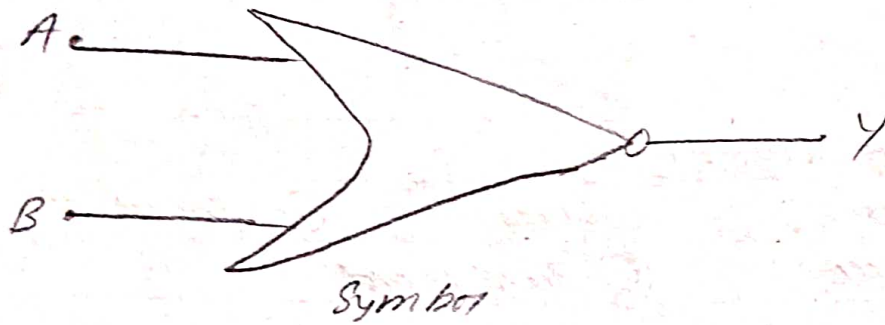


Fig: NOR gate using TTL

to the combination is equal to a given output given by the combination of NOT, AND gates. Hence, the combination of these two gives a new gate.

Boolean expression $\Rightarrow Y = \overline{A \cdot B}$

Truth Table

Inputs		Output
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

OBSERVATION:

(I) NOR Gate

Input		Output
A	B	$Y = \overline{A + B}$
0	0	1.11
0	1.28	0.05
1.28	0	0.04
1.28	1.28	0.03

(II) NAND Gate

Input		Output
A	B	$Y = \overline{A \cdot B}$
0	0	1.11
0	1.28	1.11
1.28	0	1.11
1.28	1.28	0.10

CONCLUSION:

Hence, NOR and NAND gate are verified using TTL.

PRECAUTION:

- (1) The apparatus should be connected carefully.
- (2) Voltage should be noted carefully correctly.